EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 6

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

Chapter 14: 55,57,58

55. (Using peak values)

$$e_{in} = \upsilon_a + \upsilon_b \Rightarrow \upsilon_a = e_{in} - \upsilon_b$$
$$= 60\text{ V}\angle 45° - 20\text{ V}\angle -45°$$
$$= 63.25\text{ V}\angle 63.43°$$

and $e_{in} = 63.25 \sin(377t + 63.43°)$

57. $e_{in} = \upsilon_a + \upsilon_b + \upsilon_c$

$$\upsilon_a = e_{in} - \upsilon_b - \upsilon_c$$
$$= 120\text{ V}\angle 30° - 30\text{ V}\angle 60° - 40\text{ V}\angle 120°$$
$$= 108.92\text{ V}\angle -0.33°$$

$e_{in} = \mathbf{108.92 \sin(377t - 0.33°)}$

58. $\mathbf{I}_s = \mathbf{I}_1 + \mathbf{I}_2 + \mathbf{I}_3$

$$\mathbf{I}_1 = \mathbf{I}_s - \mathbf{I}_2 - \mathbf{I}_3$$
$$= 12.73\text{ A}\angle 180° - 5.66\text{ A}\angle -180° - 2[5.66\text{ A}\angle -180°]$$
$$= 12.73\text{ A}\angle 180° - 5.66\text{ A}\angle -180° - 11.32\text{ A}\angle 180°$$
$$= 4.25\text{ A}\angle 0°$$

$i_1 = \mathbf{6.01 \sin 377t}$

Chapter 15: 14b,15b,18,30,31,39

14.

b. $\mathbf{V}_1 = \dfrac{(40\,\Omega\angle 90^\circ)(60\text{ V}\angle 5^\circ)}{6.8\,\Omega + j40\,\Omega + 22\,\Omega} = \dfrac{2400\text{ V}\angle 95^\circ}{28.8 + j40} = \mathbf{48.69\text{ V}\angle 40.75^\circ}$

$\mathbf{V}_2 = \dfrac{(22\,\Omega\angle 0^\circ)(60\text{ V}\angle 5^\circ)}{49.29\,\Omega\angle 54.25^\circ} = \dfrac{1.32\text{ kV}\angle 5^\circ}{49.29\,\Omega\angle 54.25^\circ} = \mathbf{26.78\text{ V}\angle -49.25^\circ}$

15.

b. $\mathbf{Z}_T = 4.7\text{ k}\Omega + j30\text{ k}\Omega + 3.3\text{ k}\Omega - j10\text{ k}\Omega = 8\text{ k}\Omega + j20\text{ k}\Omega = 21.541\text{ k}\Omega\angle 68.199^\circ$

$\mathbf{Z}'_T = 3.3\text{ k}\Omega + j30\text{ k}\Omega - j10\text{ k}\Omega = 3.3\text{ k}\Omega + j20\text{ k}\Omega = 20.27\text{ k}\Omega\angle 80.631^\circ$

$\mathbf{V}_1 = \dfrac{\mathbf{Z}'_T\mathbf{E}}{\mathbf{Z}_T} = \dfrac{(20.27\text{ k}\Omega\angle 80.631^\circ)(120\text{ V}\angle 0^\circ)}{21.541\text{ k}\Omega\angle 68.199^\circ} = \mathbf{112.92\text{ V}\angle 12.432^\circ}$

$\mathbf{V}_2 = \dfrac{\mathbf{Z}''_T\mathbf{E}}{\mathbf{Z}_T}$ $\qquad \mathbf{Z}''_T = 3.3\text{ k}\Omega - j10\text{ k}\Omega = 10.53\text{ k}\Omega\angle -71.737^\circ$

$= \dfrac{(10.53\text{ k}\Omega\angle -71.737^\circ)(120\text{ V}\angle 0^\circ)}{21.541\text{ k}\Omega\angle 68.199^\circ} = \mathbf{58.66\text{ V}\angle -139.94^\circ}$

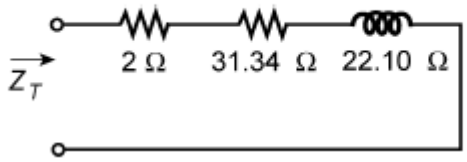
18. $P = VI\cos\theta \Rightarrow 300\text{ W} = (120\text{ V})(3\text{ A})\cos\theta$

$\cos\theta = 0.833 \Rightarrow \theta = \mathbf{33.59^\circ}$

$\mathbf{V} = 120\text{ V}\angle 0^\circ,\ \mathbf{I} = 3\text{ A}\angle -33.59^\circ$

$\mathbf{Z}_T = \dfrac{\mathbf{V}}{\mathbf{I}} = \dfrac{120\text{ V}\angle 0^\circ}{3\text{ A}\angle -33.59^\circ} = 40\,\Omega\angle 33.59^\circ = \mathbf{33.34\,\Omega + j22.10\,\Omega}$

$R_T = 33.34\,\Omega = 2\,\Omega + R \Rightarrow R = 31.34\,\Omega$



30. a. $X_L = \omega L = (1000 \text{ rad/s})(3.9 \text{ H}) = 3.9 \text{ k}\Omega$,

$$X_C = \frac{1}{\omega C} = \frac{1}{(1000 \text{ rad/s})(0.12\ \mu\text{F})} = 8.33 \text{ k}\Omega$$

$$\mathbf{Y}_T = \frac{1}{3\text{ k}\Omega\angle 0^\circ} + \frac{1}{3.9\text{ k}\Omega\angle 90^\circ} + \frac{1}{8.33\text{ k}\Omega\angle -90^\circ}$$

$$= 0.333 \text{ mS}\angle 0^\circ + 0.256 \text{ mS}\angle -90^\circ + 0.120 \text{ mS}\angle 90^\circ$$

$$= \mathbf{0.333\ mS - j0.136\ mS = 0.36\ mS\angle -22.22^\circ}$$

d. $\mathbf{E} = \mathbf{I}/\mathbf{Y}_T = 3.54 \text{ mA}\angle -20^\circ/0.36 \text{ mS}\angle -22.22^\circ = \mathbf{9.83\ V\angle 2.22^\circ}$

$$\mathbf{I}_R = \frac{E\angle\theta}{R\angle 0^\circ} = 9.83 \text{ V}\angle 2.22^\circ/3 \text{ k}\Omega\angle 0^\circ = \mathbf{3.28\ mA\angle 2.22^\circ}$$

$$\mathbf{I}_L = \frac{E\angle\theta}{X_L\angle 90^\circ} = 9.83 \text{ V}\angle 2.22^\circ/3.9 \text{ k}\Omega\angle 90^\circ = \mathbf{2.52\ mA\angle -87.78^\circ}$$

$$\mathbf{I}_C = \frac{E\angle\theta}{X_C\angle -90^\circ} = 9.83 \text{ V}\angle 2.22^\circ/8.33 \text{ k}\Omega\angle -90^\circ = \mathbf{1.18\ mA\angle 92.22^\circ}$$

g. $P = I^2R = (3.28 \text{ mA})^2 3 \text{ k}\Omega = \mathbf{32.28\ mW}$

h. $F_p = G/Y_T = 0.333 \text{ mS}/0.36 \text{ mS} = \mathbf{0.925\ leading}$

i. $e = \mathbf{13.9 \sin(1000t + 2.22^\circ)}$

$i_R \cong \mathbf{4.64 \times 10^{-3} \sin(1000t + 2.22^\circ)}$

$i_L \cong \mathbf{3.56 \times 10^{-3} \sin(1000t - 87.78^\circ)}$

$i_C = \mathbf{1.67 \times 10^{-3} \sin(1000t + 92.22^\circ)}$

31. a.

$$\mathbf{I}_1 = \frac{(60\ \Omega\angle 90^\circ)(20\text{ A}\angle 40^\circ)}{22\ \Omega + j60\ \Omega} = \frac{1200\text{ A}\angle 130^\circ}{63.91\angle 69.86^\circ} = \mathbf{18.78\angle 60.14^\circ}$$

$$\mathbf{I}_2 = \frac{(22\ \Omega\angle 0^\circ)(20\text{ A}\angle 40^\circ)}{63.91\ \Omega\angle 69.86^\circ} = \frac{440\text{ A}\angle 40^\circ}{63.91\angle 69.86^\circ} = \mathbf{6.88\angle -29.86^\circ}$$

b.

$$\mathbf{I}_1 = \frac{(12\ \Omega - j6\ \Omega)(6\text{ A}\angle 30^\circ)}{12\ \Omega - j6\ \Omega + j4\ \Omega} = \frac{(13.42\angle -26.57^\circ)(6\text{ A}\angle 30^\circ)}{12 - j2}$$

$$= \frac{80.52\text{ A}\angle 3.43^\circ}{12.17\angle -9.46^\circ} = \mathbf{6.62\ A\angle 12.89^\circ}$$

$$\mathbf{I}_2 = \frac{(4\ \Omega\angle 90^\circ)(6\text{ A}\angle 30^\circ)}{12.17\angle -9.46^\circ} = \frac{24\text{ A}\angle 120^\circ}{12.17\angle -9.46^\circ} = \mathbf{1.97\ A\angle 129.46^\circ}$$

c. $\mathbf{I}_1 = \dfrac{(-j10\ \Omega + j40\ \Omega)(4\ \text{A}\ \angle 0°)}{+j20\ \Omega - j10\ \Omega + j40\ \Omega} = \dfrac{(30\ \Omega\ \angle 90°)(4\ \text{A}\ \angle 0°)}{50\ \Omega\ \angle 90°}$

$= \mathbf{2.4\ A\ \angle 0°}$

$\mathbf{I}_2 = \dfrac{(20\ \Omega\ \angle 90°)(4\text{A}\ \angle 0°)}{50\ \Omega\ \angle 90°} = \mathbf{1.6\ A\ \angle 0°}$

39. a. $C_T = 2\ \mu\text{F}$

$X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi(10^3\ \text{Hz})(2\ \mu\text{F})} = 79.62\ \Omega$

$X_L = \omega L = 2\pi(10^3\ \text{Hz})(10\ \text{mH}) = 62.80\ \Omega$

$\mathbf{Y}_T = \dfrac{1}{220\ \Omega\ \angle 0°} + \dfrac{1}{79.62\ \Omega\ \angle -90°} + \dfrac{1}{62.8\ \Omega\ \angle 90°}$

$= 4.55\ \text{mS}\ \angle 0° + 12.56\ \text{mS}\ \angle 90° + 15.92\ \text{mS}\ \angle -90°$

$= \mathbf{4.55\ mS - j3.36\ mS = 5.66\ mS\ \angle -36.44°}$

$\mathbf{E} = \mathbf{I}/\mathbf{Y}_T = 1\ \text{A}\ \angle 0°/5.66\ \text{mS}\ \angle -36.44° = \mathbf{176.68\ V\ \angle 36.44°}$

$\mathbf{I}_R = \dfrac{E\ \angle \theta}{R\ \angle 0°} = 176.68\ \text{V}\ \angle 36.44°/220\ \Omega\ \angle 0° = \mathbf{0.803\ A\ \angle 36.44°}$

$\mathbf{I}_L = \dfrac{E\ \angle \theta}{X_L\ \angle 90°} = 176.68\ \text{V}\ \angle 36.44°/62.80\ \angle 90° = \mathbf{2.813\ A\ \angle -53.56°}$

b. $F_p = G/Y_T = 4.55\ \text{mS}/5.66\ \text{mS} = \mathbf{0.804\ lagging}$

c. $P = I^2R = (0.803\ \text{A})^2\ 220\ \Omega = \mathbf{141.86\ W}$

f. $\mathbf{I}_s = \mathbf{I}_R + 2\mathbf{I}_C + \mathbf{I}_L$

and $\mathbf{I}_C = \dfrac{\mathbf{I}_s - \mathbf{I}_R - \mathbf{I}_L}{2}$

$= \dfrac{1\ \text{A}\ \angle 0° - 0.803\ \text{A}\ \angle 36.44° - 2.813\ \text{A}\ \angle -53.56°}{2}$

$= \dfrac{1 - (0.646 + j0.477) - (1.671 - j2.263)}{2} = \dfrac{-1.317 + j1.786}{2}$

$\mathbf{I}_C = -0.657 + j0.893 = \mathbf{1.11\ A\ \angle 126.43°}$

g. $\mathbf{Z}_T = \dfrac{1}{\mathbf{Y}_T} = \dfrac{1}{5.66\ \text{mS}\ \angle -36.44°} = 176.7\ \Omega\ \angle 36.44°$

$= \mathbf{142.15\ \Omega + j104.96\ \Omega} = R + jX_L$

Chapter 16: 9,11,14

9. a. $X_{L_1} = \omega L_1 = 2\pi(10^3 \text{ Hz})(0.1 \text{ H}) = 628\ \Omega$

$X_{L_2} = \omega L_2 = 2\pi(10^3 \text{ Hz})(0.2 \text{ H}) = 1.256 \text{ k}\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi(10^3 \text{ Hz})(1\ \mu\text{F})} = 0.159 \text{ k}\Omega$$

$$\mathbf{Z}_T = R\angle 0° + X_{L_1}\angle 90° + X_C\angle -90° \parallel X_{L_2}\angle 90°$$

$$= 300\ \Omega + j628\ \Omega + 0.159 \text{ k}\Omega\angle -90° \parallel 1.256 \text{ k}\Omega\angle 90°$$

$$= 300\ \Omega + j628\ \Omega - j182\ \Omega$$

$$= \mathbf{300\ \Omega + j446\ \Omega = 537.51\ \Omega\angle 56.07°}$$

b. $$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{50 \text{ V}\angle 0°}{537.51\ \Omega\angle 56.07°} = \mathbf{93 \text{ mA}\angle -56.07°}$$

c. (CDR): $$\mathbf{I}_1 = \frac{\mathbf{Z}_{L_2}\mathbf{I}_s}{\mathbf{Z}_{L_2} + \mathbf{Z}_C} = \frac{(1.256 \text{ k}\Omega\angle 90°)(93 \text{ mA}\angle -56.07°)}{+j1.256 \text{ k}\Omega - j0.159 \text{ k}\Omega}$$

$$= \frac{116.81 \text{ mA}\angle 33.93°}{1.097\angle 90°} = \mathbf{106.48 \text{ mA}\angle -56.07°}$$

$$\mathbf{I}_2 = \frac{\mathbf{Z}_C\mathbf{I}_s}{\mathbf{Z}_{L_2} + \mathbf{Z}_C} = \frac{(0.159 \text{ k}\Omega\angle -90°)(93 \text{ mA}\angle -56.07°)}{1.097 \text{ k}\Omega\angle 90°}$$

$$= \frac{14.79 \text{ mA}\angle -146.07°}{1.097\angle 90°} = 13.48 \text{ mA}\angle -236.07°$$

$$= \mathbf{13.48 \text{ mA}\angle 123.93°}$$

d. $$\mathbf{V}_1 = (I_2\angle\theta)(X_{L_2}\angle 90°) = (13.48 \text{ mA}\angle 123.92°)(1.256 \text{ k}\Omega\angle 90°)$$

$$= \mathbf{16.93 \text{ V}\angle 213.93°}$$

$$\mathbf{V}_{ab} = \mathbf{E} - (I_s\angle\theta)(R\angle 0°) = 50 \text{ V}\angle 0° - (93 \text{ mA}\angle -56.07°)(300\ \Omega\angle 0°)$$

+-

$$= 50 \text{ V} - 27.9 \text{ V}\angle -56.07°$$

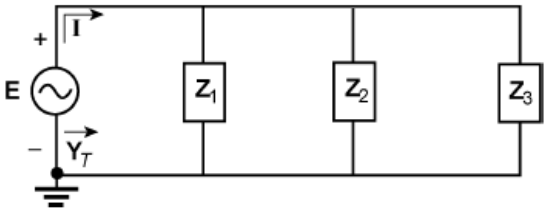
$$= 50 \text{ V} - (15.573 \text{ V} - j23.149 \text{ V})$$

$$= 34.43 \text{ V} + j23.149 \text{ V} = \mathbf{41.49 \text{ V}\angle 33.92°}$$

e. $$P = I_s^2 R = (93 \text{ mA})^2 300\ \Omega = \mathbf{2.595 \text{ W}}$$

f. $$F_p = \frac{R}{Z_T} = \frac{300\ \Omega}{537.51\ \Omega} = \mathbf{0.558 \text{ (lagging)}}$$

11. a.



$\mathbf{Z}_1 = 2\ \Omega - j2\ \Omega = 2.828\ \Omega\ \angle{-45°}$
$\mathbf{Z}_2 = 3\ \Omega - j9\ \Omega + j6\ \Omega$
$= 3\ \Omega - j3\ \Omega = 4.243\ \Omega\ \angle{-45°}$
$\mathbf{Z}_3 = 10\ \Omega\ \angle 0°$

$$\mathbf{Y}_T = \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \frac{1}{\mathbf{Z}_3} = \frac{1}{2.828\ \Omega\ \angle -45°} + \frac{1}{4.243\ \Omega\ \angle -45°} + \frac{1}{10\ \Omega\ \angle 0°}$$
$$= 0.354\ \text{S}\ \angle 45° + 0.236\ \text{S}\ \angle 45° + 0.1\ \text{S}\ \angle 0° = 0.59\ \text{S}\ \angle 45° + 0.1\ \text{S}\ \angle 0°$$
$$= 0.417\ \text{S} + j0.417\ \text{S} + 0.1\ \text{S}$$
$$\mathbf{Y}_T = 0.517\ \text{S} + j\ 0.417\ \text{S} = \mathbf{0.66\ S\ \angle 38.89°}$$
$$\mathbf{Z}_T = \frac{1}{\mathbf{Y}_T} = \frac{1}{0.66\ \text{S}\angle 38.89°} = \mathbf{1.52\ \Omega\ \angle{-38.89°}}$$

b. $$\mathbf{V}_1 = \frac{(2\ \Omega\ \angle 0°)(60\ \text{V}\angle 0°)}{2\ \Omega - j2\ \Omega} = \frac{120\ \text{V}\ \angle 0°}{2.828\ \angle -45°} = \mathbf{42.43\ V\ \angle 45°}$$

c. $$\mathbf{I}_1 = \frac{\mathbf{E}}{\mathbf{Z}'} = \frac{60\ \text{V}\ \angle 0°}{3\ \Omega - j9\ \Omega + j6\ \Omega} = \frac{60\ \text{V}\ \angle 0°}{3\ \Omega - j3\ \Omega} = \frac{60\ \text{V}\ \angle 0°}{4.243\ \angle -45°}$$
$$= \mathbf{14.14\ A\ \angle 45°}$$

d. $$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{Z}_T} = \frac{60\ \text{V}\ \angle 0°}{1.52\ \Omega\ \angle -38.89°} = \mathbf{39.47\ A\ \angle 38.89°}$$

14. $$\mathbf{Z}' = X_{C_2}\ \angle -90° \parallel R_1\ \angle 0° = 2\ \Omega\ \angle{-90°} \parallel 1\ \Omega\ \angle 0°$$
$$= \frac{2\ \Omega\ \angle -90°}{1 - j2} = \frac{2\ \Omega\ \angle -90°}{2.236\ \angle -63.435°}$$
$$= 0.894\ \Omega\ \angle{-26.565°}$$
$$\mathbf{Z}'' = X_{L_2}\ \angle 90° + \mathbf{Z}' = +j8\ \Omega + 0.894\ \Omega\ \angle{-26.565°}$$
$$= +j8\ \Omega + (0.8\ \Omega - j4\ \Omega)$$
$$= 0.8\ \Omega + j4 = 4.079\ \Omega\ \angle 78.69°$$

$$\mathbf{I}_{X_{L_2}} = \frac{X_{C_1}\angle -90°\mathbf{I}}{X_{C_1}\angle -90° + \mathbf{Z}''} = \frac{(2\ \Omega\angle -90°)(0.5\ \text{A}\ \angle 0°)}{-j2\ \Omega + (0.8\ \Omega + j4\ \Omega)} = \frac{1\ \text{A}\ \angle -90°}{0.8 + j2}$$
$$= \frac{1\ \text{A}\ \angle -90°}{2.154\ \angle 68.199°} = 0.464\ \text{A}\ \angle -158.99°$$
$$\mathbf{I}_1 = \frac{X_{C_2}\ \angle -90°\ \mathbf{I}_{X_{C_2}}}{X_{C_2}\angle -90° + R_1} = \frac{(2\ \Omega\ \angle -90°)(0.464\ \text{A}\ \angle -158.99°)}{-j2\ \Omega + 1\Omega} = \frac{0.928\ \text{A}\ \angle -248.99°}{2.236\ \angle -63.435°}$$
$$= \mathbf{0.42\ A\ \angle 174.45°}$$